

Topological Social Manifolds: Contractibility, Echo Chambers, and Inverse Centrality Optimization Paradoxes

System Architecture Invariant Manifest

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1 The Ecclesial Social Manifold Layer

To bridge the gap between God's atemporal control laws (W_2) and the material double-integrator machine fleet (W_1), we formalize the human collective network layer as a geometric manifold with strict algebraic topology constraints.

1.1 Step 1: Manifolds God Accepts (Trivial Homology)

The primitive Church complex represents a contractible social manifold with trivial higher-dimensional homology ($H_n(X) = 0$ for $n > 0$). This guarantees that the entire population complex can continuously collapse via the Morse vector field down to a single global 0-simplex consensus anchor: Jesus Christ ($\beta_1 = 0, \beta_0 = 1$). The network implements three definitive algorithmic interaction policies:

1. **Radical Edge-Weight Equalization (Galatians 3:28):** Strips away real-world biases to eliminate un-collapsible sub-cliques.
2. **The Matthew 18 Low-Pass Friction Filter:** Confines disputes within local sub-simplices, executing step-by-step dimensional collapses to trap emotional noise before it triggers global network panic.
3. **Resource Risk Pooling (Acts 2:44):** Standardizes cross-node asset margins to dramatically minimize individual node failure rates.

1.2 The Alternative Optimization Paradox

Rather than maximizing traditional metrics like PageRank or Degree Centrality, the manifold enforces an Inverse Centrality Reward. Node utility is inversely proportional to hub hierarchy:

$$\mathcal{R}_{\text{hub}} \propto \frac{1}{\text{Degree}(v)} \quad (1)$$

requiring high-degree nodes (leaders) to expend energy supporting peripheral leaf nodes. The network compensates agents with non-zero-sum intrinsic fruit payouts (love, joy, peace), driving the individual's existential fear parameter to zero ($\theta_i \rightarrow 0$).

1.3 Step 2: Manifolds God Rejects (The Twelve Spies Counterexample)

When collective decision-making fails, it triggers a topological tear. By bypassing structural filtration nodes, the 10 unfaithful spies generate a horizontal shortcut edge. This creates a non-trivial 1-dimensional homology cycle ($H_1 \neq 0$):

$$H_1(\mathcal{G}_{\text{spies}}) \neq 0 \implies \beta_1 \geq 1 \quad (2)$$

The population gets trapped inside an un-collapsible algorithmic echo chamber, circulating panic infinitely around the space-time loop, rendering the target destination V' structurally unobtainable and forcing an unrecoverable mission abort.